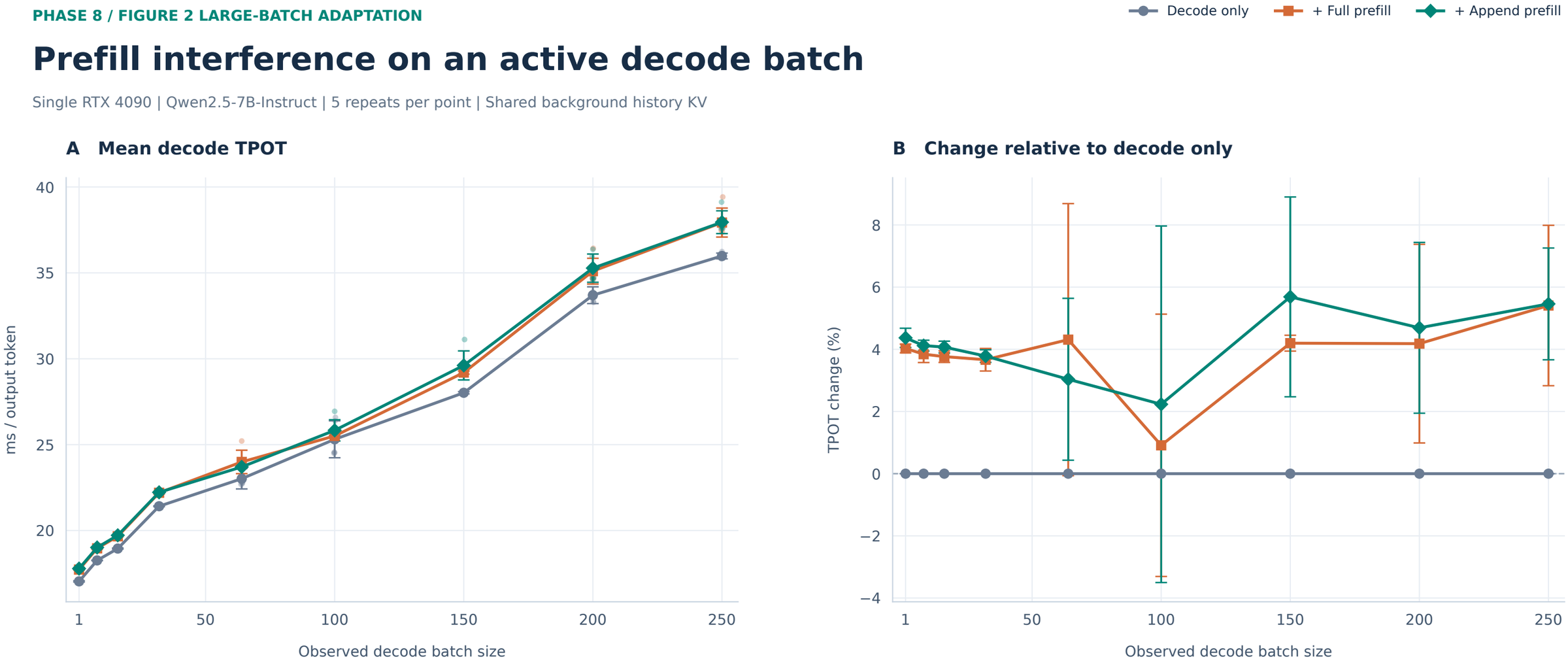
PHASE 8 / FIGURE 2 LARGE-BATCH ADAPTATION

Prefill interference on an active decode batch

Single RTX 4090 | Qwen2.5-7B-Instruct | 5 repeats per point | Shared background history KV



Mean of trial means; error bars = sample SD across 5 repeats. Faint dots = individual trials. Curves use the actual measured batch sizes.
Full: 0 cached + 1024 new tokens. Append: 2048 cached + 1024 new tokens. Decode: 2048 cached + 4 new, then 128 output tokens.
Injection occurs after every decode emits >=8 tokens. Scheduler evidence verifies actual batch size and mixed execution; no KV network transfer is involved.
This is a hardware/model adaptation of Figure 2, not the original H100/Llama result. TPOT averages cover the full decode stream; one-step effects are diluted.
Background sequences share the same 2048-token history KV, with distinct 4-token suffixes and separate generation. This is not independent-KV scaling.
Observer overhead: P99 0.20 ms, max 229.30 ms. Occasional stalls are retained; small differences are not significance claims.